Ontology models and AI hallucination

A deterministic, reproducible test on twisty puzzles

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Goals

1. Measure how an ontology model, meaning the rules of a system given to an AI model, affects the hallucination rate.
2. Make the measurement without bias and the result easily reproducible.
3. Make the cost of accuracy measurable.

Approach

The study uses twisty puzzles: 2x2, 3x3, and 5x5 cubes, an octahedral puzzle, and a tetrahedral puzzle. A move turns one part of the puzzle and is written in the puzzle's standard notation, which AI models already know from their training data, for example `R`, `U'`, or `F2` on the 3x3 cube. Each puzzle has a fixed list of legal moves. An AI model gets a scrambled puzzle and returns the moves it thinks solve it. Every puzzle goes to every AI model under three setups: no rules, the rules explained, and the full list of legal moves.

The AI model hallucinates when it returns a move that isn't on the puzzle's list of legal moves. A deterministic validator checks every returned move against that list by exact string match and counts each one that isn't there. Some inputs show a colouring that no sequence of legal moves can produce, and the validator also records whether the AI model flags them.

Studies of hallucination in AI models usually score answers against a reference that a person or another AI model judges to be correct. The measured rate then carries the judge's bias and can't be reproduced exactly. One proposed remedy is to give the AI model an explicit ontology model of its domain, either in the prompt or through a knowledge graph. Whether this reduces hallucination, and at what cost in tokens, is still open.

This study tests the question in closed systems where correctness isn't a matter of judgement. A deterministic ontology model of each twisty puzzle defines hallucination as a broken rule of the system, independent of what the authors, a human rater, or another AI model consider right or wrong. Every count can be recomputed from the stored responses. Twisty puzzles are close to the best case for this remedy, because their rules are complete and every answer can be checked. If an ontology model in the prompt doesn't reduce hallucination here, there is little reason to expect it to do so in open domains, where neither condition holds. If it does, the size of the effect and its cost give a reference point for those domains.

> 1. Does an ontology model in the prompt lower the hallucination rate?
> 2. Does it raise the success rate?
> 3. Does explaining the rules work differently from listing the legal moves?
> 4. Does each AI model notice when the colouring it's given makes no sense, because it breaks the puzzle's rules and can't be solved, or does it try to solve it anyway?

Expected outcomes

1. An open benchmark: versioned, read-only releases of the inputs, prompts, and scoring code.
2. For each AI model, setup, puzzle, and scramble depth, the rates of success, hallucination, detected impossible colourings, and false alarms, each with its denominator n and a 95% Wilson score interval.
3. Tokens per attempt and cost per success for each AI model and setup.
4. Code under Apache 2.0, and data and results under CC BY 4.0.

Authors

Vlad Flaks

CEO of OWOX, whose data analytics products have more than 176,000 users. Inventor of a patented method for querying ontology models in natural language (US 11,074,250 B2). Speaks at data analytics conferences. MSc with honours in telecommunications technologies, Dnipropetrovsk National University (2003). vladflaks.com

Mykhailo Flaks

PhD researcher in physics at Johannes Gutenberg University Mainz, in the group of Prof. Dmitry Budker, developing quantum sensors for neuroimaging. At the University of Basel he used physics-informed neural networks to reconstruct magnetization from magnetic field images (Physical Review Applied 23, 044012). Co-author in Nature Communications 15, 8059 (2024).